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24 June 2026

L05 Chihuahua or Muffin with CNN: Reflective Journal

This laboratory provided a practical introduction to Convolutional Neural Networks (CNNs) and demonstrated why they have become one of the most important architectures in computer vision. The objective of the activity was to build, train, and evaluate a CNN capable of distinguishing between images of Chihuahuas and muffins. Although the classification task appears simple, it highlighted several important concepts in deep learning and allowed me to compare CNNs with the traditional neural network model developed in the previous assignment.

The CNN architecture used in this lab consisted of three convolutional layers followed by ReLU activation functions and max-pooling layers. After feature extraction, the resulting feature maps were flattened and passed through fully connected layers for classification. This architecture differs significantly from the traditional neural network used in Modulo 04. In the previous lab, images were flattened into a one-dimensional vector before being processed by fully connected layers. As discussed in the course materials, this approach ignores the spatial relationships between pixels and requires a large number of parameters. CNNs address this limitation by using convolutional filters that examine local image regions, share weights across the image, and progressively learn hierarchical features such as edges, shapes, textures, and objects. This design allows CNNs to preserve spatial information while reducing the number of parameters needed for image recognition tasks.

The model achieved a validation accuracy of 96.67%, which indicates strong performance on the Chihuahua-versus-muffin classification task. During evaluation, most images were classified correctly, with only a small number of misclassifications. One particularly interesting example was a Chihuahua image that was incorrectly predicted as a muffin. This error illustrates an important challenge in computer vision: some objects may share visual characteristics that confuse a model. In this case, the Chihuahua's coloration, facial features, or image composition may have resembled patterns learned from muffin images. Despite this isolated mistake, the model demonstrated a high degree of accuracy and generalization on the validation dataset.

Comparing the CNN to the traditional neural network from the previous workshop revealed clear advantages. In Lab 04, the traditional neural network achieved approximately 80% validation accuracy. In contrast, the CNN achieved 96.67% accuracy on the same classification problem. This improvement supports the idea presented in the course materials that CNNs are specifically designed for image data because they exploit spatial structure and local patterns. The trade-off is that the CNN required a longer training process and a more complex architecture. The traditional neural network was conceptually easier to understand because it consisted primarily of fully connected layers, while the CNN introduced additional concepts such as convolutions, filters, feature maps, pooling operations, and data augmentation. Nevertheless, the performance improvement justified the increased complexity.

One of the main challenges I faced during this laboratory was understanding the complete workflow of a CNN. Compared to previous labs, the code was significantly more complex and included many components working together simultaneously. In addition to defining the neural network architecture, it was necessary to understand data transformations, image resizing, normalization, augmentation, dataloaders, loss functions, optimizers, and training loops. Another challenge was interpreting the architecture summary and understanding how image dimensions changed throughout the network. I overcame these difficulties by carefully reviewing the lecture materials and connecting the code implementation to the theoretical concepts discussed in the module. Observing the training progress, validation accuracy, and prediction results also helped reinforce my understanding of how CNNs learn visual patterns.

This type of image classification model has many practical real-world applications. CNNs are widely used in medical imaging to detect diseases from X-rays, MRIs, and CT scans. They are also employed in autonomous vehicles for object detection, traffic sign recognition, and pedestrian identification. Other applications include quality control in manufacturing, facial recognition systems, wildlife monitoring, security surveillance, and content moderation on digital platforms. The ability of CNNs to automatically learn visual features makes them highly effective for tasks involving large collections of image data.

Despite their effectiveness, CNN-based systems raise several ethical considerations. One concern is dataset bias. If the training data are not representative of real-world conditions, the model may perform poorly for certain groups or scenarios. Another concern is the impact of misclassification. In low-risk applications such as distinguishing between Chihuahuas and muffins, an incorrect prediction may have minimal consequences. However, in domains such as healthcare, transportation, or security, classification errors could lead to serious outcomes. Privacy is another important issue, particularly when image recognition systems process personal photographs or surveillance data. For these reasons, human oversight, transparency, and responsible data collection practices remain essential when deploying computer vision systems in real-world environments.

Overall, this laboratory helped me understand why CNNs became the dominant architecture in computer vision. By comparing the results of the CNN with those of a traditional neural network, I gained a clearer understanding of how convolutional layers improve image classification performance. The experience reinforced the importance of selecting an architecture that matches the structure of the data and demonstrated how deep learning models can automatically learn meaningful visual features from images. While the implementation was more challenging than previous labs, it provided valuable insight into the foundations of modern computer vision systems.

**References**


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